

Warning: Theorem 4 is about a coefficient matrix, not an augmented matrix. If an augmented matrix $[A \ b]$ has a pivot position in every row, then the equation $Ax=b$ may or may not be consistent.

Computation of Ax

The calculations in Example 1 were based on the definition of the product of a matrix A and a vector x . The following simple example will lead to a more efficient method for calculating the entries in Ax when working problems hand by hand.

Example 4: Compute Ax , where $A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix}$ and $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Solution: From the definition,

$$\begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 5 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ -3 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 2x_1 \\ -x_1 \\ 6x_1 \end{bmatrix} + \begin{bmatrix} 3x_2 \\ 5x_2 \\ -2x_2 \end{bmatrix} + \begin{bmatrix} 4x_3 \\ -3x_3 \\ 8x_3 \end{bmatrix} \quad \text{(Scalar multiplication)}$$

2
$$= \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ -x_1 + 5x_2 - 3x_3 \\ 6x_1 - 2x_2 + 8x_3 \end{bmatrix} \quad \text{(Vector addition)}$$

The first entry in the product Ax is a sum of products (sometimes called a dot product), using the first row of A and the entries in x .

3 That is,
$$\begin{bmatrix} 2 & 3 & 4 \\ -1 & 5 & -3 \\ 6 & -2 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 + 3x_2 + 4x_3 \\ \vdots \\ \vdots \end{bmatrix}$$

This matrix shows how to compute the first entry in Ax directly, without writing down all the calculations shown in (1). Similarly, the second entry in Ax can be calculated at once by multiplying the entries in the second row of A by the corresponding entries in x .